CHAPTER 27

# Proposal Writing

*Proposals are written offers to initiate a proposed course of action.*

## LEARNING OBJECTIVES

- Understanding the nature and importance of proposals
- Knowing the different types of proposals
- Learning about the components of formal proposals
- Understanding the proposal writing process

## 27.1 INTRODUCTION

Suppose that the new sales manager of a company finds that his company is facing stiff competition from rivals and sales have been decreasing for the last several months. He has a brilliant idea to increase the sales of his company and tells it to his sales director. The sales director is enthused and asks the sales manager to provide a 'proposal'. What he wants is that the sales manager should write about his 'brilliant idea' and provide a 'rationale' for it.

> A proposal is a systematic, factual, formal, and persuasive description of a course of action or set of recommendations or suggestions.

Most people find it very difficult to write a proposal that persuades the readers to accept the proposed course of action. The reason is simple. The given proposal may contain a course of action or set of suggestions but the 'rationale' is either absent or is too weak to be considered. No one is going to accept what a proposal states unless they are convinced about its viability. Thus, it is essential to be persuasive in order to write an effective proposal.

A proposal is a method of persuading people to agree to the writers view or accept his suggestions. It is a systematic, factual, formal, and persuasive description of a course of action or set of recommendations or suggestions. It is written for a specific audience to meet a specific need. As the main objective of a proposal is to persuade the reader to accept the proposed course of action, it explains and justifies what it proposes. Engineers, scientists, researchers, business executives, managers, and administrators have to write proposals in order to initiate new projects, provide fresh ideas, solve problems, or reinforce and prompt innovative strategies.

> Proposals initiate new projects, provide fresh ideas, solve problems, or reinforce and prompt innovative strategies.

Submitting a proposal is usually the first step in going ahead with a new project. Whether an agency needs to be persuaded to work on a project, a business deal has to be initiated with a company, or a potential customer needs to be persuaded to purchase goods/services, a proposal may need to be submitted first. The proposal may be accepted or rejected depending on how effectively it responds to the needs of the situation/problem or the institution/company for whom the proposal is prepared. It is important that the proposal convinces the reader/s that the proposed course of action will lead to future benefits by showing an understanding of readers' needs and offering a viable way to fulfil those needs.

## 27.2 TYPES OF PROPOSALS

As summarised in Table 27.1, proposals can be classified as **non-formal** and **formal,** according to structure, as **internal** and **external** according to the nature of its audience, and as **solicited** and **unsolicited** according to the source of origin. Each of these have been discussed briefly.

### 27.2.1 Non-Formal and Formal Proposals

Proposals can be formal or non-formal depending on their content and format. A non-formal proposal is a brief description of suggestions or recommendations that are introductory in nature. It is usually written to initiate small projects that do not require elaborate description and discussion. Non-formal proposals are usually short. As the content is generally insufficient for a formal proposal, a non-formal proposal may involve the use of printed forms, letter formats, or memo formats.

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Formal proposals, on the other hand, are comparatively longer. They are usually written to initiate big projects and require elaborate description and discussion. Like a formal report, a formal proposal involves use of the manuscript format. It may consist of several sections and sub-sections and can vary from a few pages to hundreds of pages.

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**TABLE 27.1 Types of Proposals**

| *Criteria* | *Types* | *Description* |
|---|---|---|
| Content and format | Non-formal<br>Formal | Short proposals involving small projects<br>Long proposals with elaborate description and discussion |
| Nature of audience | Internal<br>External | Addressed to readers within an organisation<br>Communicated to people outside an organisation |
| Source of origin | Solicited<br>Unsolicited | Written in response to a request for proposal<br>Written without any request for proposal |

## 27.2.2 Internal and External Proposals

Proposals can be internal or external, according to the nature of the audience. An internal proposal is addressed to readers within an organisation. It may offer to study a problem, situation, condition, or issue in the company or organisation, and may present different options for solving it. For example, proposing a plan to increase the sales of a company will require preparing an internal proposal. Internal proposals are less formal and elaborate than external proposals.

External proposals are communicated to people outside an organisation. An external proposal may offer a plan to solve a problem or situation of another organisation and give appropriate suggestions and recommendations. External proposals are more formal, detailed, and elaborate than internal proposals.

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### *27.2.3 Solicited and Unsolicited Proposals*

Proposals may be classified into two types: solicited or unsolicited. A solicited proposal is written in response to a specific request from a client. Many companies, government agencies, institutions, and consultancy organisations solicit proposals for their projects. As they want the best people to take their projects, they may make the request for proposal open to increase competition. They specify their requirements and mention their conditions.

In contrast, unsolicited proposals are written without any request for a proposal. As they intend to propose solutions or recommendations, they are based on an objective assessment of a situation or condition by an individual or a firm. For example, a person noticing a problem in his organisation and wanting to offer his ideas on how to handle it, may submit an unsolicited proposal. Self initiated research and business projects usually involve unsolicited proposals.

## Progress Check 1

**1. Which of the following statements are true in the light of the above discussion about the nature, significance and types of proposals?**

(a) Proposals are oral offers to solve problems.
(b) All proposals are internal.
(c) Proposals never tell how a situation can be handled.
(d) A proposal is a method of persuading other people to agree to your views or accept your suggestions.
(e) Most projects may begin with proposals.
(f) Non-formal proposals may be written in letter or memorandum form.
(g) Professionals do not have to write unsolicited proposals.
(h) A proposal is a formal document written for a specific audience to meet a specific need.
(i) Proposals reinforce, prompt, motivate, or persuade the readers to act.
(j) The most important purpose of a proposal is to describe a situation as it is.

## 27.3 WRITING EFFECTIVE PROPOSALS

### 27.3.1 Structure of Formal Proposals

The structure of a formal proposal is similar to that of a formal report. The structure of a formal proposal is discussed here.

> The structure of a formal proposal is similar to that of a formal report.

#### *Parts of a Formal Proposal*

A formal proposal may include some or all the following parts:

#### *Title Page*

The title page of a proposal contains the title of the proposal, the name of the person or organisation to whom the proposal is being submitted, the name of the proposal writer, and the date (Fig. 27.1).

A proposal on
--------------------------------------------------
Submitted to
--------------------------------------------------
Submitted by
--------------------------------------------------
Date
--------------------------------------------------

**Fig. 27.1** Structure of Title Page

### *Table of Contents*

This section provides the reader an overall view of the proposal by listing the main headings and the sub-headings in the proposal, with their page numbers (Fig. 27.2).



**Fig. 27.2** Structure of Table of Contents

### *List of Figures*

This section includes a list of tables, graphs, figures, and charts used in the proposal, with their page number/s (Fig. 27.3).



**Fig. 27.3** Structure of List of Figures

### *Abstract or Summary*

An abstract or a summary is a condensed version of the proposal as it summarises and highlights its major points. However, an abstract is more specialised and technical than an executive summary.

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### *Methodology*

The section on methodology summarises the proposed methods of data collection and the procedure for investigating the situation/problem.

### *Introduction*

This section introduces readers to the introduction to the proposal. It gives the background, states the purpose, and discusses the scope. It may also try to persuade readers by highlighting the major advantages and justifying the proposed course of action.

### *Statement of the Problem*

This section contains an objective description of the problem or situation that the proposal intends to address. As it links the proposed course of action to the needs of the reader and the requirements of the situation, it gives credibility to the proposal and makes it convincing and acceptable.

### *Proposed Plan and Schedule*

This section presents a schedule of activities, highlighting the main course of action.

### *Advantages/Disadvantages*

This section reinforces that the proposal has more advantages than disadvantages by making realistic comparisons. It links benefits to the needs of the situation.

### *Recommendations/Proposed Solutions*

This is the main section of a proposal as it discusses the plan to solve the problem. It is the most persuasive section of a proposal. It is usually the longest section of a proposal, and is logically structured into small manageable sub-sections with suitable headings.

### *Conclusion*

This section presents the final summary of the proposal and focuses on the main points, and the key benefits. It influences readers with a final appeal.

### *Appendices*

Secondary materials are put as appendices in a proposal. This maintains continuity of logical progression and avoids distractions in the main text of the proposal.

## Progress Check 2

**1. Study the following table and match different components of a formal proposal (right column) with their functions (left column):**

| | Functions | | Components of a Formal Proposal |
|---|---|---|---|
| 1. | Describes the proposed methods of data collection and the procedures for investigating the situation/problem | (a) | Abstract or summary |
| 2. | Contains secondary material | (b) | Methodology |
| 3. | Contains an objective description of the problem or situation that the proposal intends to address | (c) | Introduction |
| 4. | Links benefits to the needs of the situation | (d) | Statement of problem |
| 5. | Discusses the plan to solve the problem | (e) | Proposed plan and schedule |
| 6. | Presents a schedule of activities highlighting the main course of action | (f) | Advantages/Disadvantages |
| 7. | Presents the final summary of the proposal and focuses on the main points | (g) | Recommendation |
| 8. | Summarises and highlights the major points of a proposal | (h) | Conclusions |
| 9. | Introduces the proposal by presenting its major aspects | (i) | Appendices |

## 27.3.2 Writing Strategies

Apart from using the correct format and structure for the proposal, the proposal should be readable, attractive, and convincing. In order to take any action, the reader/s should be able to understand the proposal. If the proposal is confusing, complex, or too abstract, the reader will not be able to respond to it positively. So, simple and appropriate language should be used to make the proposal readable. Make the proposal attractive and convincing so that reader/s can take a positive decision after reading it. In order to achieve these objectives, a systematic plan of writing should be followed and strategies of good writing adopted.

> Proposal writing is persuasive writing that should involve prewriting, writing, and post-writing.

### *Pre-Writing*

Prewriting of a proposal involves purpose identification, audience analysis, project analysis, scope determination, an analysis of the action desired, and data collection. The writing process should begin with the following questions:

1. Why is this proposal being written?/What are its objectives?
   (*Purpose identification*)
2. Who is the audience?
   (*Audience analysis*)
3. Does the proposal involve any project? What is the project?
   *(Project analysis)*
4. How much information should be included in the proposal?
   (*Scope determination*)
5. What should the reader do?
   (*Analysis of the action desired*)

Once these five questions are answered, the writer can collect data related to the proposal. The writer should do background research, collect relevant information, discuss relevant points with concerned people, make a list of the points that he/she wants to cover, and organise his/her thoughts to help him/her write.

### *Writing*

Writing a proposal involves organising the data that has been collected, outlining what will be presented in the proposal, and writing the first draft. The information may be organised as per the structure of the proposal. A good outline will give the writer a clear picture of his/her proposal and will help him/her to prepare the first draft.

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### *Post-Writing*

Once the first draft has been written, it is ready to be revised, edited, and evaluated in order to improve its content, layout, and structure. The proposal needs to be edited for grammatical and lexical accuracy. The first draft should be evaluated and critically examined to ensure that the proposal can achieve its purpose. Finally, the final draft is prepared.

### 27.3.3 Sample Proposal

The following sample shows the structure, components, layout, and style of a proposal. Please note that only some sections of the full proposal have been included here.

**TITLE PAGE**

A PROPOSAL ON
Development of Erbium Doped Fibre Amplifiers for
Wavelength Division Multiplexed Optical Communication System

SUBMITTED TO
Ministry of Human Resource Development
Bureau of Technical Education
Government of India

SUBMITTED BY
Dr. Vishnu Priye
Department of Electronics and Instrumentation
Indian School of Mines, Dhanbad – 826 004, Jharkhand

OCTOBER 3, 2013

**ABSTRACT**

The proposed work is experimental in nature with software to be developed based on equations to analyse Erbium doped fibre amplifiers (EDFAs) that are available in literature and also developed by the Principal Investigator as a part of a previous R&D Project on ASE Broadband Optical Source sponsored by the Ministry of Information and Communication Technology, New Delhi.
The main objectives of the proposed project are:

- to fabricate EDFAs having gain ≥ 20 dB, noise figure ≤ 5 dB, and gain flattened for WDM applications in the conventional C-Band using fibres and other components commercially available in the international market,
- to fabricate EDFAs having above specifications, using indigenously developed components at CGCRI, Kolkata, CEERI Pilani, IIT Delhi, and Optiwave Photonics, Hyderabad to make it cost effective for the Indian market, and
- to develop software to estimate the performance and characteristic features of EDFA.

**METHODOLOGY**

The methodology to be adopted will be as follows:

- Different pumping schemes, erbium doped fibres (EDFs) of different doping levels, and different doping configurations such as clad doped EDFs, different lengths, and other physical parameters will be experimentally investigated to achieve the target of accomplishing optical gain ≥ 20 dB and noise figure ≤ 5 dB at one signal wavelength (1550 nm). For the experiments, commercial as well as indigenous components will be used.

- Using a tunable laser source, the gain spectrum of the above EDFA in the C-Band (1525 – 1565 nm) will be optimised so as to give equal gain and noise figure in the whole wavelength range so that it can be applied in Wavelength Division Multiplexed systems.
  The stipulated period of completion of the project is two years.
- Using wavelength multiplexed signals the response of the module will be investigated for real system application.

## INTRODUCTION

**Importance of the area of study**
EDFAs have assumed importance because they provide format independent gain and have replaced the more expensive and limited electronic regenerators. They are responsible for recent advances in optical communication systems for long haul links. For fibre optic communication systems, they can be used as power amplifiers to boost transmitter power, as in-line amplifiers to increase the system non-repeater reach, or as pre-amplifiers to enhance receiver sensitivity.

**Manpower requirement**
Manpower is required to carry out experiments, compile results, and write technical details. As the Principal Investigator has teaching load, research load and administrative responsibilities one Project Associate with minimum qualification of BTech/MSc on a consolidated monthly salary of Rs 10,000/- is essential.

**Nature of ongoing activities pertaining to project under submission at the Institution.**
The Principal Investigator has recently completed an R&D Project titled “Design of Tunable Broadband ASE Fiber Source and Multiwavelength Fibre Laser for WDM System” sponsored by the Ministry of Information and Communication Technology (MICT), New Delhi, which resulted in a Broadband ASE tunable source in C-Band and a Software based on Genetic Algorithm that has been copyrighted by MICT. He is also involved in designing optical modulators, isolators, polarisers, optical networks as R&D, PhD programmes and as B Tech projects. He has recently obtained a minor research project from ISM Dhanbad on Design of—Wavelength—Routed Optical Networks Using Genetic Algorithm. One PhD student is working with him on the application of EDFA to optical networks.

**Existing facilities concerned with the project under reference**

(a) *Level of infrastructure available with reference to technology under consideration*
At present, the Fibre Optics Laboratory of the Electronic & Instrumentation Department, ISM Dhanbad, has essential equipment such as optical spectrum analyser, splicing machine, 980 nm laser diode modules, wavelength selective couplers, isolators, and few metres of EDF that can be used immediately to start the project in the gestation period of purchase of new equipment/components.

(b) *Library facilities*
Departmental and Central Library facilities are available. However, for successful running of the project, specific technical books have to be procured, which relate to the broad area of optical amplifier and communication.

(c) *Workshop facilities are available*

**Linkage with sister institutions, research laboratories, industry, and other agencies**

There will be association with CGCRI, Kolkata, CEERI, Pilani, and IIT Delhi in terms of technical discussions, characterisation of modules and visits to one another's laboratories. However no financial obligations will be there. No MOU has been signed.

**Self assessment reflecting specific competence for undertaking the project**

The Principal Investigator has been involved in research and developmental activities in the area of Fibre and integrated optics' for the past 18 years. During this period, he has developed various analytical and numerical techniques to model optical fibres and integrated optical waveguides, as well as fibre optic components, including isolators, polarisers, polarisation splitters and so on. He is principal investigator of the recently completed R&D project on ASE Broadband Optical Source using EDF sponsored by Ministry of Information and Communication Technology (MICT), New Delhi, which resulted in a Broadband ASE tunable source in C-Band, and a software based on genetic algorithm, which has been copyrighted by MICT.

## STATEMENT OF THE PROBLEM

Optical fibre amplifiers with broadband amplification have become important with the continuously increasing number of wavelength channels in optical network systems. Erbium doped fibres (EDFs) are an integral part of present state-of the art optical amplifiers. EDFs consist of silica fibres doped with optically active rare earth erbium ($Er^{3+}$) ions (typical doping level ~200 mole ppm). The energy levels of the erbium ions in the silica glass host shows excited energy levels corresponding to wavelengths of interest 1530 nm, 980 nm and 800 nm. When a laser beam corresponding to the wavelength of 980 nm is launched in the EDF, the $Er^{3+}$ ions at ground level get excited to upper levels. For $Er^{3+}$ ions in glass host all transitions are non-radiative except between the levels corresponding to wavelength 1530 nm. Hence, most $Er^{3+}$ ions relax down to this level from where they undergo, either, spontaneous or stimulated emission to the ground level. Due to stark split energy levels, the emission occurs in the wavelength range of 1400–1600 nm with relative flat wavelength response in the range 1525–1565 nm.

When EDF ($Er^{3+}$ doping 180 mole ppm) of a given length (~ 12-15 m) is pumped by a 980 nm laser of about 80 mW power, the whole length of fibre is totally population inverted. If a signal at a wavelength in the range 1525 – 1565 nm is launched, it induces stimulated emission and as a result the signal gets amplified. Apart from stimulated emission, there are spontaneous emissions in this wavelength range due to finite lifetime of the atoms in excited states. The photons emitted spontaneously also get amplified and add to the noise of the amplifier.

Presently, international research is devoted to investigating different means to increase the gain and reduce the noise figure of EDFAs by investigating different pumping schemes; different doping levels; different dopants such as ytterbium and phosphate along with erbium; as well as different dopant configuration such as cladding doping, depressed core, and others. The gain flattening is also being investigated so that all signals have equal gain in wavelength multiplexed systems. Schemes are being proposed to reduce cross talks and the effect of non-linearity in fibres.

Research is being done at IIT Delhi as part of the PhD programme and also as research projects. Till 24th March 1999, the principal investigator as a Senior Scientist at IIT Delhi was involved in a Technology Development Mission (TDM) project funded by the MHRD and Planning Commission, towards the technology development of a prototype model of an erbium doped fibre amplifier (EDFA). C-Dot is also involved in development and characterisation of EDFAs. CGCRI, Kolkata, is developing EDFs; CEERI, Pilani, is developing 980 nm Laser diodes; and IIT Delhi is developing wavelength selective couplers for application in EDFA.

**PROPOSED BUDGET**

**A. NON-RECURRING**

| **Sl. No.** | **Equipment/Components** | **Estimated cost In Lakhs (INR)** |
|---|---|---|
| 1. | 980 nm Laser Diode Module (Imported) | 3.0 |
| 2. | 1480 nm Laser Diode Module (Indigenous) | 1.0 |
| 3. | Tunable laser (Range 1500 –1600 nm) (To be imported) | 6.0 |
| 4. | DFB Laser (1550 nm) module (Indigenous) | 2.0 |
| 5. | Multiplexer/Demultiplexer (Imported) | 3.0 |
| 6. | Erbium Doped Fibres (Both indigenous and imported) | 3.0 |
| 7. | Wavelength Selective Couplers (Imported) | 1.6 |
| 8. | Tap Couplers (Imported) | 1.0 |
| 9. | *Fiber Bragg Grating* | 2.0 |
| **A.** | **Total (Non-Recurring): (Rupees Twenty Two lakhs Sixty Thousand Only)** | **22.60** |

**B. RECURRING**

| **S. No.** | **Head** | **Estimated Cost In Lakhs (INR)** |
|---|---|---|
| 1. | Consumables | 0.30 |
| 2. | Travel/Training | 0.40 |
| 3. | Contingencies (Books + journals + communication services etc.) | 0.50 |
| **B.** | **Total (Recurring): (Rupees One Lakh Twenty Thousand Only)** | 1.20 |

Total cost of the proposed project (A+B):₹ 23.80 (Rupees Twenty three lakhs eighty thousand only)

## PROPOSED PLAN

**Plan of Execution of Project Under Consideration**

**Organisational set up**

The Institute has an Assistant Registrar (Projects) whose responsibility it is to ascertain the financial transparency related to the project.

**Principal Coordinator**

Dr Vishnu Priye, Assistant Professor, Department of Electronics and Instrumentation, ISM Dhanbad will be the chief coordinator of the project.

**Suggested Plan of action for development of infrastructure**

The existing rules of the Institute for purchase will be followed. The acquisition of permanent equipment, systems, and sub-systems will be taken up at the earliest for enhancing the pace of the project. Orders will be placed for required books and journals related to the area of Optical Amplifiers and Optical Communications.

**Plan for internal monitoring and evaluation of progress of implementation**

For verifying the progress of the proposed project the Institute will take up internal monitoring. Every six months evaluation of the progress of the project can be made by a intra-institute committee. The progress report will be sent to the funding agency as well at regular intervals. The Institute has an officer for auditing the project. It can be further audited by the external audit.

**Stipulated period of completion**

The stipulated period of completion of the project is two years.

This is a modified version of the original proposal submitted to the Ministry of Human Resource Development of the Government of India by Dr. Vishnu Priye, Assistant Professor, Department of Electronics and Instrumentation, Indian School of Mines, Dhanbad. The proposal has since been approved by the GOI. We are thankful to Dr. Vishnu Priye for permiting us to reproduce the proposal in a modified form.

## Exercise

1. **Write short notes on the following:**
   (a) Nature and significance of proposals
   (b) Formal and non-formal proposals
   (c) Solicited and unsolicited proposals
   (d) External and internal proposals
2. **Discuss the different components of a formal proposal.**
3. **Read the sample proposal given in this chapter and rewrite the proposal in the form of a letter, excluding unnecessary details.**

## Key to Progress Check

### *Progress Check 1*

1. (d), (e), (f), (h), (i)

### *Progress Check 2*

1. 1 (b), 2 (i), 3 (d), 4 (f), 5 (g), 6 (e), 7 (h), 8 (a), 9 (c)